Partial Fractions

Questions 1–32

Question 1. Express $\dfrac{7x+17}{(x+1)(x+3)}$ in the form $\dfrac{A}{x+1}+\dfrac{B}{x+3}$.

Question 2. Express $\dfrac{12x-3}{(x-2)(x+1)}$ in the form $\dfrac{A}{x-2}+\dfrac{B}{x+1}$.

Question 3. Write each of the following as partial fractions:

a) $\frac{2x+3}{(x-2)(x+5)}$

b) $\frac{16x+1}{(x+4)(2x+1)}$

c) $\frac{4}{(x+2)(x+7)}$

d) $\frac{14}{(x+2)(x+6)}$

e) $\frac{4}{x(x-2)}$

f) $\frac{x+9}{x^2+8x+15}$

g) $\frac{49-x}{2x^2+13x-24}$

h) $\frac{2(11x+4)}{8x^2+6x+1}$

i) $\frac{k}{x^2+x-20}$

where k is a positive constant.

Question 4. Tom is working on a partial fraction question and so far he has written:

$$\frac{1}{25-4x^2} \equiv \frac{A}{5-2x} + \frac{B}{5+2x}$$
$$25-4x^2 \equiv A(5+2x) + B(5-2x)$$
$$\text{When } x = -\frac{5}{2},\ 0 = -10B \Rightarrow B = 0.$$

a) State what his mistake is.

b) Find the correct partial fraction.

Question 5. Express $\dfrac{3x^2-14x+31}{(x-5)(x-3)(x+1)}$ in the form $\dfrac{A}{x-5}+\dfrac{B}{x+1}+\dfrac{C}{x-3}$.

Question 6. Write each of the following as partial fractions:

a) $\frac{4x^2-25x+61}{(x-5)(x-2)(x+1)}$

b) $\frac{2}{(x-4)(x+3)(2x-1)}$

Question 7. Given $f(x) = 4x^3 - 8x^2 - 15x + 9$

a) show that $(2x - 1)$ is a factor of $f(x)$

b) Hence fully factorise, $f(x)$

c) Write the following as partial fractions $\dfrac{5}{4x^3 - 8x^2 - 15x + 9}$.

Question 8. Lisa is working on a partial fraction question and so far she has written:

$$\frac{1}{(x+2)(x^2-1)} \equiv \frac{A}{x+2} + \frac{B}{x-1} + \frac{C}{x+1}$$
$$1 \equiv A(x+2) + B(x-1) + C(x+1)$$
$$\text{When } x = -2,\ 1 = -3B - C$$
$$\text{When } x = 1,\ 1 = 3A + 2C$$
$$\text{When } x = -1,\ 1 = A - 2B$$

a) State what her mistake is.

b) Find the correct partial fractions.

Question 9. Express in the form $\dfrac{7x^2+21x+12}{(x+1)^2(x+3)}$ in the form $\dfrac{A}{x+1}+\dfrac{B}{(x+1)^2}+\dfrac{C}{x+3}$.

Question 10. Express in the form $\dfrac{32-5x-x^2}{(x-2)^2(x+4)}$ in the form $\dfrac{A}{x-2}+\dfrac{B}{(x-2)^2}+\dfrac{C}{x+4}$.

Question 11. Write each of the following as partial fractions:

a) $\frac{3}{(x-1)(x+2)^2}$

b) $\frac{2x+1}{(2x-1)(x+1)^2}$

c) $\frac{2}{x^2(x-1)}$

d) $\frac{x+3}{(x-1)^2}$

Question 12. For each of the following select the format you should use to begin the method of finding the partial fraction:

A: $\dfrac{A}{x+2}+\dfrac{B}{x-3}$

B: $\dfrac{A}{x+2}+\dfrac{B}{(x+2)^2}+\dfrac{C}{x-3}$

C: $\dfrac{A}{x+2}+\dfrac{B}{x-3}+\dfrac{C}{(x-3)^2}$

a) $\frac{5}{(x+2)(x-3)}$

b) $\frac{4x+1}{(x+2)^2(x-3)}$

c) $\frac{3x-2}{x^2-x-6}$

d) $\frac{2x-1}{x^3+x^2-8x-12}$

e) $\frac{5x^2+1}{x^3-4x^2-3x+18}$

Question 13. Write each of the following as partial fractions:

a) $\frac{2x+3}{x^2+3x-10}$

b) $\frac{4}{x^3-4x^2+4x}$

c) $\frac{5x+1}{4x^3+8x^2-x-2}$

d) $\frac{2}{2x^3+x^2-4x-3}$

Question 14. Given $f(x) = \dfrac{5x-1}{(x-2)(x+7)}$, $x \neq 2$, $x \neq -7$.

a) Write $f(x)$ as partial fractions.

Given $g(x) = \dfrac{3x^2-19x+44}{(x+7)(x-2)^2}$, $x \neq 2$, $x \neq -7$.

b) Write $g(x)$ as partial fractions.

c) Hence, find the exact solution to the equation $f(x) = g(x)$.

Question 15. It is known that $\dfrac{5x+19}{(x+3)(x+p)^2} \equiv \dfrac{1}{x+3} - \dfrac{1}{x+p} + \dfrac{3}{(x+p)^2}$ for some value of p. Find the value of p.

Question 16. Prerequisite knowledge for Q16–Q19: Differentiation of $(ax+k)^n$ Given $f(x) = \dfrac{x+2}{(x-4)(2x+1)}$, $x \neq 4$, $x \neq -\frac{1}{2}$.

a) Write $f(x)$ as partial fractions.

b) Find the exact coordinates of the stationary points of the curve $y = f(x)$.

Question 17. Given $f(x) = \dfrac{3x}{(x-2)^2(5-x)}$, $x \neq 2$, $x \neq 5$.

a) Write $f(x)$ as partial fractions.

b) Show that the tangent to the curve $y = f(x)$ at the point $x = 3$ has the equation $4y + 21x = 81$.

Question 18. Given $f(x) = \dfrac{2x+1}{(x+3)(2-x)}$, $x > -3$.

a) Write $f(x)$ as partial fractions.

b) Show that $f(x)$ is always increasing.

Question 19. Given $f(x) = \dfrac{5}{(x+1)(x+2)}$, $x < -1$.

a) Write $f(x)$ as partial fractions.

b) Show that $f(x)$ is always decreasing.

Question 20. Prerequisite knowledge for Q20–Q22: Binomial Series Given $f(x) = \dfrac{1}{(1-2x)(1+x)}$, $x \neq \frac{1}{2}, x \neq -1$.

a) Write $f(x)$ as partial fractions.

b) Hence expand $f(x)$ up the term in x^3.

c) State the values for which the expansion is valid.

Question 21. Given $f(x) = \dfrac{1}{(x+1)(x-2)}$, $x \neq -1, x \neq 2$.

a) Write $f(x)$ as partial fractions.

b) Hence expand $f(x)$ up the term in x^3.

c) State the values for which the expansion is valid.

Question 22. Given $f(x) = \dfrac{3x^2 - 5x + 6}{(1-x)^2(3+x)}$, $x \neq 1, x \neq -3$.

a) Write $f(x)$ as partial fractions.

b) Hence expand $f(x)$ up the term in x^3.

c) State the values for which the expansion is valid.

Question 23. Given $\dfrac{8x^2+4x+1}{(2x-1)(x^2+1)} \equiv \dfrac{A}{2x-1} + \dfrac{Bx+C}{x^2+1}$ find the values of A, B and C.

Question 24. The following questions consider improper fractions Given $\dfrac{3x+2}{x-1} \equiv A + \dfrac{B}{x-1}$.

a) find the values of A and B

b) Hence, sketch $y = \dfrac{3x+2}{x-1}$.

Question 25. Given $\dfrac{12-5x}{2-x} \equiv A + \dfrac{B}{x-2}$.

a) find the values of A and B

b) Hence, sketch $y = \dfrac{12-5x}{2-x}$.

Question 26. Find the constants in the following identities:

a) $10 - 4x^2 - 7x \equiv (Ax + B)(1 - 4x) + C$

b) $2x^2 + 15 \equiv (Ax + B)(3 - x) + C$

c) $3x^3 + 2x^2 \equiv (Ax + B)(x^2 + 1) + Cx + D$

d) $x^3 + 2 \equiv (Ax^2 + Bx + C)(x + 1) + Dx$

Question 27. Given $\dfrac{x^2-3x-13}{x-5} \equiv A + Bx + \dfrac{C}{x-5}$ find the values of A, B and C.

Question 28. Given $\dfrac{2+5x-5x^2}{1-x} \equiv Ax + \dfrac{B}{1-x}$ find the values of A and B.

Question 29. Given $\dfrac{x^3}{x^2+3x-4} \equiv A + Bx + \dfrac{C}{x-1} + \dfrac{D}{x+4}$ find the values of A, B and C.

Question 30. Given $\dfrac{x^4-1}{(x-2)(x-1)(x+4)} \equiv Ax + B + \dfrac{C}{x-2} + \dfrac{D}{x+4}$ find the values of A, B, C and D.

Question 31. Given $\dfrac{2x^4 - x^3 - x + 4}{x^2 + 1} \equiv Ax^2 + Bx + C + \dfrac{D}{x^2 + 1}$ find the values of A, B, C and D.

Question 32. Prerequisite knowledge for Q32: Arithmetic Series

a) Write $\dfrac{4p + 5}{(p + 1)(p + 2)}$, $p \neq -1$, $p \neq -2$ in partial fractions.

It is known that $\dfrac{A}{p + 1}$ and $\dfrac{B}{p + 2}$ are the first two terms in an arithmetic sequence.

b) Using your values of A and B find the third term of the arithmetic sequence in terms of p.

The sum of the first 3 terms is 1.

c) Find the value of p and state the first 3 terms of the sequence.